

Mohammad Mahmudul Hasan, PhD

Associate Professor

Department of Electrical and Electronic Engineering



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EDUCATION

- Norwegian University of Science and Technology, Norway** (05.2022 – 06.2025)
Doctor of Philosophy
Information and Communication Technology
- KIIT University, India** (07.2008 – 06.2010)
Master of Technology
Communication Systems Engineering
- KIIT University, India** (07.2004 – 06.2008)
Bachelor of Technology
Electronics and Telecommunication Engineering

WORK EXPERIENCE

- University of Information Technology and Science, Bangladesh** (01.2011 – Present)
 - ✓ Associate Professor, Electrical and Electronic Engineering 07.2025 – Present
 - ✓ Assistant Professor, Electrical and Electronic Engineering 01.2011 – 06.2025
 - ✓ Advisor, Information and Communication Technology Cell 01.2018 – 04.2022
 - ✓ Head of the Department, Electronics and Communication Engineering 02.2019 – 04.2022
 - ✓ Director, Information and Communication Technology Cell 12.2020 – 04.2022
 - ✓ Project Manager, Enterprise Resource Planning 03.2020 – 04.2022
 - ✓ Editorial Board Member, Journal of Science and Engineering 10.2020 – 04.2022
 - ✓ Member, Self-Assessment Committee, IQAC 07.2017 – 07.2018
 - ✓ Member, Finance Committee 03.2020 – 04.2022
 - ✓ Member, Academic Council 02.2019 – 04.2022
- Brno University of Technology, Czech Republic** (08.2024 – 01.2025)
 - ✓ Visiting Research Fellow, Department of Automation
- Norwegian University of Science and Technology, Norway** (05.2022 – 06.2025)
 - ✓ Research Fellow, Faculty of Engineering
- Sanyo Engineering & Construction Inc., Japan** (04.2018 – 01.2019)
 - ✓ Industrial Trainer, BJIT Limited, Bangladesh
- KIIT University, India** (02.2009 – 12.2010)
 - ✓ Assistant Professor, Electronics and Telecommunication Engineering 07.2010 – 12.2010
 - ✓ Teaching Assistant, Electronics and Telecommunication Engineering 02.2009 – 06.2010

SELECTED PUBLICATIONS

- Inaamullah Khan, **Mohammad Mahmudul Hasan**, Michael Cheffena Gebresilassie, “Transmitter-Assisted Joint Data-Aided Channel Estimation and PAPR Reduction Scheme in Wireless Fading Channels”, **Scientific Reports**, vol. 16, no. 1, 8015, 2026 (2026)
- Mohammad Mahmudul Hasan**, Onur Alev, Pavel Skrabanek, Gabriela Soukupová, Fatima Hassouna, and Michael Cheffena Gebresilassie, “Microwave MIMO E-Nose for Wireless Communication and Selective Detection of VOC Mixtures with Concentration Estimation”, **ACS Sensors**, vol. 10, no. 9, pp. 6446–6463, 2025 (2025)
- Inaamullah Khan, **Mohammad Mahmudul Hasan**, Michael Cheffena Gebresilassie, “A Novel Low-Complexity Peak-Power-Assisted Data-Aided Channel Estimation Scheme for MIMO-OFDM Wireless Systems”, **IEEE Open Journal of Signal Processing**, vol. 6, pp. 992 – 1003, 2025 (2025)
- Onur Alev, **Mohammad Mahmudul Hasan**, Emel Tuğba Ertuğrul, Selçuk Birdoğan, Okan Özdemir, Eda Goldenberg, Michael Cheffena, “Hydrothermally Synthesized Molybdenum disulfide Nanoflakes: Structural, Electrical, and Antenna-based Gas Sensing Characteristics”, **Sensors & Actuators: A. Physical**, vol. 393, 116756, 2025 (2025)

1. **Mohammad Mahmudul Hasan**, Onur Alev, Pavel Skrabanek, and Michael Cheffena Gebresilassie, "Molecularly Imprinted Polymer-Based Electronic Nose for Ultrasensitive, Selective Detection and Concentration Estimation of VOC Mixtures", **IEEE Sensors Journal**, vol. 25, no. 10, 2025 (2025)
2. **Mohammad Mahmudul Hasan**, Todd Cowen, Onur Alev and Michael Cheffena Gebresilassie, "MIMO Microwave Sensor for Selective and Simultaneous Detection of Methanol and Ethanol Gases at Room Temperature," **IEEE Transactions on Instrumentation & Measurement**, vol. 74, 9511613, 2025 (2025)
3. **Mohammad Mahmudul Hasan**, Onur Alev and Michael Cheffena Gebresilassie, "Dual-Functional Antenna Sensor for Highly Sensitive and Selective Detection of Isopropanol Gas Using Optimized Molecularly Imprinted Polymers," **ACS Sensors**, vol. 10, no. 3, pp. 2147–2161, 2025 (2025)
4. **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena Gebresilassie, "MoS₂/MoO_x Nanoflake-Based Dual-Functional Antenna Sensors for Highly Sensitive and Selective Detection of Volatile Organic Compounds," **ACS Applied Nano Materials**, vol. 7, no. 21, pp. 25065–25077, 2024 (2024)
5. **Mohammad Mahmudul Hasan**, Todd Cowen and Michael Cheffena Gebresilassie, "A Novel Molecularly Imprinted Polymer-Based Carbon Nanotube-Coated Microwave Sensor for Selective Detection of Methanol Gas," **IEEE Sensors Letters**, vol. 8, no. 5, 6004904, 2024 (2024)
6. **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena Gebresilassie, "A Novel Molybdenum Disulfide-Based High-Precision Microwave Sensor for Methanol Gas Detection at Room Temperature," **IEEE Microwave and Wireless Technology Letters**, vol. 34, no. 6, pp. 691 – 694, 2024 (2024)
7. **Mohammad Mahmudul Hasan**, Michael Cheffena Gebresilassie, "Adaptive Antenna Impedance Matching Using Low-Complexity Shallow Learning Model", **IEEE Access**, vol. 11, pp. 74101 – 74111, 2023 (2023)
8. **Mohammad Mahmudul Hasan**, Michael Cheffena Gebresilassie, Slobodan Petrovic, "Physical-layer Security Improvement in MIMO OFDM Systems using Multilevel Chaotic Encryption", **IEEE Access**, vol. 11, pp. 64468 – 64475, 2023 (2023)
9. Inaamullah Khan, Michael Cheffena Gebresilassie, **Mohammad Mahmudul Hasan**, "Data Aided Channel Estimation for MIMO-OFDM Wireless Systems Using Reliable Carriers", **IEEE Access**, vol. 11, pp. 47836 – 47847, 2023 (2023)
10. **Mohammad Mahmudul Hasan**, Mohammad Mahadi Hasan Foad, "Modified Gamma Correction Companding for PAPR Reduction in OFDM Systems Considering Solid State Power Amplifier and Wireless Channels", **Circuits, Systems, and Signal Processing**, vol. 37, no. 10, pp. 4431- 4454, 2018 (2018)
11. **Mohammad Mahmudul Hasan**, Mohammad Mahdi Hasan Faisal, "IGCC for PAPR Reduction in OFDM Systems over the Nonlinearity of SSPA and Wireless Fading Channels", **Circuits, Systems, and Signal Processing**, vol. 35, no. 8, pp. 2855–2880, 2015 (2015)
12. **Mohammad Mahmudul Hasan**, "PAR Reduction in SU/MU-MIMO OFDM Systems using OPF Precoding over the Nonlinearity of SSPA", **Wireless Personal Communications**, vol. 83, no. 3, pp. 2225-2248, 2015 (2015)
13. **Mohammad Mahmudul Hasan**, "A Novel CVM Precoding Scheme for PAPR Reduction in OFDM Transmissions", **Wireless Network**, vol. 20, no. 6, pp. 1573-1581, 2014 (2014)
14. **Mohammad Mahmudul Hasan**, "A New PAPR Reduction Scheme for OFDM Systems Based on Gamma Correction", **Circuits, Systems, and Signal Processing**, vol. 33, no. 5, pp. 1655-1668, 2014 (2014)
15. **Mohammad Mahmudul Hasan**, "A New PAPR Reduction Technique in OFDM Systems Using Linear Predictive Coding", **Wireless Personal Communications**, vol. 75, no. 1, pp. 707-721, 2014 (2014)
16. **Mohammad Mahmudul Hasan**, "VLM Precoded SLM Technique for PAPR Reduction in OFDM Systems", **Wireless Personal Communications**, vol. 73, no. 3, pp. 791-801, 2013 (2013)
17. **Mohammad Mahmudul Hasan**, "PAPR Reduction in OFDM Systems Based on Autoregressive Filtering", **Circuits, Systems, and Signal Processing**, vol. 33, no. 5, pp. 1637-1654, 2013 (2013)

RESEARCH PROJECTS

- Machine Learning Assisted Signal Processing for Antenna-Sensor Array (MLSP-ASA). (2024 – 2025)
Member, Research Council of Norway (Project Number: 353890)
- Air Quality Monitoring Exploiting Massive Multiple-Input Multiple-Output Antennas (AQMA). Member, Research Council of Norway (Project Number: 324061) (2022 – 2025)

PROFESSIONAL CERTIFICATIONS

- Brno University of Technology, Czech Republic** (08.2024 – 01.2025)
Professional Competence, Electrical Engineering – NV No. 194/2022 Coll.
- GrameenPhone Ltd., Bangladesh** (07.2007 – 10.2007)
Intern Engineer, Transmission Planning Division
- All India Radio & Doordarshan (Prasar Bharati), India** (05.2007 – 07.2007)
Industrial Training, Broadcasting Corporation of India
- Red Hat Bhubaneswar, India** (04.2006 – 07.2006)
Industrial Training, Red Hat Linux RHEL 4

GRANTS & AWARDS

- Research Recognition** (06.2025)
Nominated for *Best Doctoral Thesis* in Sensors
Norwegian University of Science and Technology
- Research Grant** (08.2024)
Awarded 146 000 NOK
The Research Council of Norway
- Chancellor's Gold Medal** (12.2010)
Awarded for *securing the highest CGPA* (10/10)
KIIT University, India
- Founder's Gold Medal** (12.2010)
Awarded for securing the *first position* in Master of Technology
KIIT University, India
- Chancellor's Merit Scholarship** (2004 – 2010)
Awarded for *academic excellence*, Bachelor's and Master's studies
KIIT University, India.

LANGUAGE PROFICIENCY

- ☑ Bengali (native), English (fluent), Hindi (fluent), Arabic (A1), and Norsk (A2/B1)

CITIZENSHIP & RESIDENCY

- 🇬🇧 Bangladeshi (Citizen), Norway (Permanent Residency)

REFERENCES

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I declare that the above information is true and correct to the best of my knowledge.

– Mohammad Mahmudul Hasan